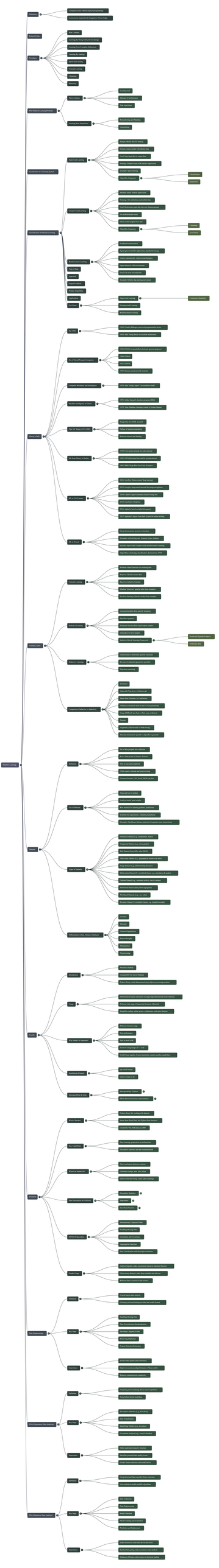

- Machine Learning
  - Definition
    - Computers learn without explicit programming
    - Autonomous acquisition & integration of knowledge
  - Related Fields
  - Paradigms
    - Rote Learning
    - Learning By Being Told (Advice-taking)
    - Learning From Examples (Induction)
    - Learning By Analogy
    - Speed-up Learning
    - Concept Learning
    - Clustering
    - Discovery
  - Well Defined Learning Problems
    - Three Features
      - Learning task
      - Measure of performance
      - Task experience
    - Learning from Experience
      - Remembering and Adapting
      - Generalizing
  - Architecture of a Learning System
  - Classification of Machine Learning
    - Supervised Learning
      - Sample labeled data for training
      - System creates model with labeled data
      - Goal: Map input data to output data
      - Analogy: Student learns with teacher supervision
      - Example: Spam filtering
      - Algorithm Categories
        - Classification
        - Regression
    - Unsupervised Learning
      - Machine learns without supervision
      - Training with unlabeled, unclassified data
      - Goal: Restructure input data into new features/groups
      - No predetermined result
      - Finds useful insights from data
      - Algorithm Categories
        - Clustering
        - Association
    - Reinforcement Learning
      - Feedback-based method
      - Agent gets reward for right action, penalty for wrong
      - Learns automatically, improves performance
      - Agent interacts with environment
      - Goal: Get most reward points
      - Example: Robotic dog learning movement
    - Type of Data
    - Approach
    - Output Feedback
    - Popular Algorithms
    - Applications
    - Use Cases
      - Supervised Learning
        - Continuous Quantities
      - Unsupervised Learning
      - Reinforcement Learning
  - History of ML
    - Pre-1940
      - 1834: Charles Babbage conceived programmable device
      - 1936: Alan Turing theory on machine instructions
    - Era of Stored Program Computers
      - 1940: ENIAC invented (first electronic general-purpose)
      - 1949: EDSAC
      - 1951: EDVAC
      - 1943: Human neural network modeled
    - Computer Machinery and Intelligence
      - 1950: Alan Turing's paper 'Can machines think?'
    - Machine Intelligence in Games
      - 1952: Arthur Samuel's checkers program (IBM)
      - 1959: Term 'Machine Learning' coined by Arthur Samuel
    - First 'AI' Winter (1974-1980)
      - Tough time for AI/ML research
      - Failure of machine translation
      - Reduced interest and funding
    - ML from Theory to Reality
      - 1959: First neural network for echo removal
      - 1985: NETtalk (neural network for pronunciation)
      - 1997: IBM's Deep Blue beat Garry Kasparov
    - ML at 21st Century
      - 2006: Geoffrey Hinton coined 'deep learning'
      - 2012: Google's deep neural network for image recognition
      - 2014: Chabot Eugen Goostman cleared Turing Test
      - 2014: Facebook's DeepFace
      - 2016: AlphaGo beat Lee Sedol (Go game)
      - 2017: Alphabet's Jigsaw team built system for online trolling
    - ML at Present
      - Great advancement, present everywhere
      - Examples: self-driving cars, Amazon Alexa, chatbots
      - Includes Supervised, Unsupervised, Reinforcement Learning
      - Algorithms: clustering, classification, decision tree, SVM
  - Learning Types
    - Concept Learning
      - Boolean-valued function over training data
      - Purpose: Classify unseen data
      - Based on inductive learning
      - Machine 'discovers' general rules from examples
      - Involves learning a function (rule) from examples
    - Inductive Learning
      - General principles from specific instances
      - Specific to general
      - Estimates function from input/output samples
      - Generalize for new samples
      - Inductive Bias & Learning Framework
        - Restricted Hypothesis Space
        - Preference Bias
    - Deductive Learning
      - General rules to determine specific outcomes
      - Reverse of induction (general to specific)
      - Top-down reasoning
    - Comparison (Deductive vs Inductive)
      - Definition
      - Approach (Top-down vs Bottom-up)
      - Starts from (Premises vs Conclusion)
      - Validity (Conclusion must be true vs Not guaranteed)
      - Usage (Difficult, true facts vs Fast, easy, evidence)
      - Process
      - Argument (Valid/invalid vs Weak/strong)
      - Structure (General to specific vs Specific to general)
  - Datasets
    - Definition
      - Set of data grouped into collection
      - Rows: Data points, Columns: Features
      - Vary in size and complexity
      - Often require cleaning and preprocessing
      - Common formats: CSV, Excel, JSON, zip files
    - Use of Datasets
      - Train and test AI models
      - Analyze trends, gain insights
      - Raw material for learning patterns, predictions
      - Essential for experiments, validating hypotheses
      - Examples: Healthcare (disease patterns), Companies (user interactions)
    - Types of Datasets
      - Numerical Dataset (e.g., temperature, marks)
      - Categorical Dataset (e.g., color, gender)
      - Web Dataset (from APIs, often JSON)
      - Time series Dataset (e.g., geographical terrain over time)
      - Image Dataset (e.g., differentiating diseases)
      - Multivariate Dataset (2+ correlated classes, e.g., attendance & grades)
      - Ordered Dataset (e.g., customer reviews, movie ratings)
      - Partitioned Dataset (data points segregated)
      - File-Based Datasets (e.g., .csv, .xlsx)
      - Bivariate Dataset (2 correlated classes, e.g., height & weight)
    - Differentiation (Data, Dataset, Database)
      - Contains
      - Structure
      - Context/Organization
      - Purpose/Insights
      - Manipulation
      - Preprocessing
  - NumPy
    - Introduction
      - Numerical Python
      - Created 2005 by Travis Oliphant
      - Python library: multi-dimensional array objects, processing routines
    - Usage
      - Mathematical/logical operations on large high-dimensional arrays/matrices
      - Perform wide range of numerical functions efficiently
      - Simplifies coding, online access, collaborates with other libraries
    - Why NumPy is Important?
      - Reduced memory usage
      - Fast performance
      - Easy to work with
      - Tools for integrating C/C++ code
      - Useful linear algebra, Fourier transform, random number capabilities
    - Installation & Import
      - pip install numpy
      - import numpy as np
    - Interoperability & Array
      - Interoperability Features
      - Multi-dimensional array representation
  - PANDAS
    - What is Pandas?
      - Python library for working with datasets
      - Name from 'Panel Data' and 'Python Data Analysis'
      - Created by Wes McKinney in 2008
    - Key Capabilities
      - Data cleaning, preparation, transformation
      - Descriptive statistics and data summarization
    - What Can Pandas Do?
      - Find correlations between columns
      - Calculate average, max, min values
      - Delete irrelevant/wrong values (data cleaning)
    - Data Description in PANDAS
      - Descriptive Statistics
      - Importance
      - describe() Function
    - PANDAS Specialties
      - Summarizing Categorical Data
      - Handling Missing Data
      - Correlation and Covariance
      - Aggregation Functions
      - Data Visualization with Descriptive Statistics
    - Pandas Usage
      - Analyze big data, make conclusions based on statistical theories
      - Clean messy datasets, make them readable and relevant
      - Relevant data is crucial in data science
  - Data Preprocessing
    - Definition
      - Crucial step in data analysis
      - Cleaning and transforming raw data into usable format
    - Key Steps
      - Handling Missing Data
      - Data Normalization/Standardization
      - Encoding Categorical Data
      - Removing Duplicates
      - Feature Selection/Extraction
    - Importance
      - Ensures data quality and consistency
      - Improves accuracy and performance of data models
      - Reduces computational complexity
  - EDA (Exploratory Data Analysis)
    - Definition
      - Analyzing and visualizing data to uncover patterns
      - Done before formal modeling
    - Key Steps
      - Descriptive Statistics (e.g., describe())
      - Data Visualization
      - Identifying Outliers (e.g., box plots)
      - Correlation Analysis (e.g., corr() in Pandas)
    - Importance
      - Helps understand dataset's structure
      - Identifies potential data quality issues
      - Guides feature selection and model choice
  - PDA (Predictive Data Analysis)
    - Definition
      - Using historical data to predict future outcomes
      - Uses statistical models and ML algorithms
    - Key Steps
      - Data Collection
      - Data Preprocessing
      - Model Selection
      - Model Training and Evaluation
      - Prediction and Deployment
    - Importance
      - Helps businesses make data-driven decisions
      - Enables forecasting, risk assessment, trend analysis
      - Enhances efficiency and accuracy in decision-making